YEAR 9 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION 10.10 - Ohm's Law

Perform the following investigation about the relationship between voltage and resistance.

Use a light globe and a ceramic resistor - do two investigations and have two tables of results.

Change the voltage from 2 V to 12 V - record your data as you go.

INVESTIGATION 10.10

Changing resistance

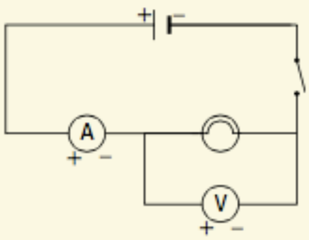
AIM **To investigate the relationship between voltage and resistance of a light globe**

Materials:

power supply (variable)
9-volt light globe and holder
6 connecting leads with alligator clips or banana plugs
switch
ammeter
voltmeter

METHOD AND RESULTS

- Set up the circuit shown in the diagram on the right and leave the switch open.



1. Construct a table like the one below in which to record your measurements of the electric current flowing through the light globe, the voltage drop across the globe and the calculated resistance.

- Set the power supply to 2 volts.

2. Close the switch and quickly read the meters, recording the electric current and voltage drop in your table. Ensure that the electric current is recorded in amperes (not milliamperes).

- With the switch closed, set the power supply to 4 volts.

3. Quickly measure and record the electric current and voltage displayed on the meters.
4. Set the power supply to 6 volts (again not opening the switch) and quickly measure and record the electric current and voltage displayed on the meters.
5. Calculate the resistance of the globe for each of the three power supply settings and record them in your table.
6. Plot a graph of voltage drop (*V*) versus electric current (*I*) for the light globe.
7. Repeat the experiment, constructing a new table in which to record your data. This time, however, start with the power supply set to 6 volts. Then decrease it to 4 volts and then 2 volts.
8. Plot a second graph of the voltage drop versus electric current on the same set of axes as the first graph in a different colour.

DISCUSS AND EXPLAIN

9. Does the resistance increase or decrease during the first part of the experiment, when the power supply setting is being increased?
10. How is the change in resistance different during the second part of the experiment, when the power supply setting is being decreased?
11. What changing property of the filament do you think caused the resistance of the light globe to change?
12. Explain any difference between the shape of the first graph and that of the second graph.

The characteristics of a light globe

Power supply setting (volts)	Electric current (amperes)	Voltage drop (volts)	Resistance (ohms)
2			
4			
6			